Krishna Tripathi

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Education

Indian Institute of Information Technology, Nagpur December 2021 to May 2025
Bachelor of Technology in Computer Science and Engineering *Nagpur, India*

Technical Skills

Languages: Python, Go, JavaScript (ES6+), TypeScript, C++, Java, SQL, Rust (Basics)
Frontend: React, Next.js, Redux Toolkit, TanStack Query, HTML5, CSS, SCSS, Figma **AI/LLM**: GenAI, Claude, LangGraph, MCP, RAG, Langfuse
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, Microservices, MCP
Testing: Jest, JUnit, Cypress, Go unit and HTTP testing, Echo (Labstack)
Databases: PostgreSQL, YugabyteDB, MongoDB, BigQuery, pgvector
Cloud & Web Engineering: AWS S3, SQS, Lambda, Cognito, GCS, Docker, Jenkins, Linux, HTTP Caching, CORS, XSS/CSRF Prevention, Lighthouse

Professional Experience

Yugabyte Bengaluru, India
Software Engineer I *September 2025 to Present*

- **Replaced polling-based file ingestion with event-driven processing** using AWS S3 **ObjectCreated events**, **SQS**, and **Lambda**; implemented presigned S3 uploads across 3 services with file-size, type, and spoofing guardrails.
- **Configured Meko's React, Go, Python, and Nginx environment**, API routing, and Langfuse observability; wrote **100+ UI/backend unit tests** with Cypress and Echo (Labstack) for Go handlers, middleware, and endpoints.
- **Built retrieval observability and context-promotion workflows** across memory and knowledge-base sources, replacing full conversation replay and reducing input-token usage by **35%** in marketing and lead-generation workflows.
- **Implemented AWS Cognito authentication across Google, LinkedIn, and GitHub**, streamlining federated identity linking, profile completion, and invite/logout flows to reduce signup time by **23%**; used TanStack Query for server-state management and implemented browser caching and web-security measures.
- **Built Python-based RAG retrieval for YugabyteDB** through a `PgDistRagRetriever` for pg_dist_rag, implementing metadata filtering, hybrid search, and integration tests; developed Java-based embedding-store and chat-memory integrations for LangChain4j, contributing code and documentation upstream in a merged commit.
- **Built reusable React UI components with Redux Toolkit for client-state management** and optimized disaster-recovery workflows backed by Go services, reducing node-selection latency from **450 ms to 120 ms**; surfaced C++ replication-lag metrics through Prometheus.
- **Developed a database migration validator across 5+ repositories**, checking schema versions and dirty state to detect invalid migrations before merge and provide actionable error feedback.
- **Strengthened YugabyteDB's open-source developer ecosystem** with JavaScript driver logging, ORM load-balancing examples, integration guidance, and JUnit tests for Yugabyted UI backend services.

Software Engineer Intern *September 2024 to August 2025*

- **Recreated the YB Voyager workflow in Yugabyted UI with UX designers using Figma** across 4 migration stages; added status and progress analytics and redesigned the migration assessment UX. (PR #2564 | implementation)
- **Built Go backend APIs and integration workflows across 5 external systems**: Scarf, GCS, BigQuery, MadKudu, and Salesforce, supporting centralized growth metrics, lead scoring, and CRM synchronization.
- **Streamlined yb-voyager developer tooling across 3 workflows**: added Homebrew install tracking, fixed Go/Docker setup in Jenkins helpers, standardized build-script versioning, corrected assessment HTML, and moved stable-version tests to Jenkins. (PR #2652 | PR #2779)
- **Improved YugabyteDB onboarding** by documenting TypeORM smart-driver setup and fixing Linux/RHEL quick-start links. (PR #28250 | PR #28418)

Zluri Bengaluru, India
Software Engineer Intern *August 2024 to September 2024*

- **Resolved 13+ production bugs** in JavaScript/Node.js and MongoDB; built QA dashboards and a testing framework, increasing test coverage by **20%** and improving query performance by **30%**.

Projects

EternalAgni: Solar Timing & Reminder App July 2026 to Present
C++17, Android NDK, CMake, JavaScript, Capacitor, PWA ***WEB*** | ***GOOGLE PLAY***

- **Engineered a native C++17 astronomical timing engine** using Android NDK and CMake, generating sunrise and sunset schedules with ±1-second precision (tested against source) and supporting 92 days of offline operation.
- **Built reliable cross-platform scheduling** across Android and PWA, implementing exact alarms, boot-time recovery, service-worker caching, duplicate suppression, and protection against late or missed reminders.
- **Optimized native Android deployment** with architecture-specific shared libraries, 16 KB ELF page-size alignment, and R8 code shrinking and obfuscation for modern Android compatibility.

YB Voyager Copilot: AI-Powered Database Migration Assistant January 2025 to Present
LangGraph, FastAPI, Next.js, Python, MCP, Docker, YugabyteDB, RAG ***DOCKER HUB*** | ***ARCHITECTURE*** | ***DEMO***

- **Architected a 3-agent LangGraph and MCP workflow** spanning all 4 migration stages and grounded in product documentation.
- **Built a 3-tier RAG backend and containerized 5 services** with FastAPI, pgvector, YugabyteDB, and Docker; automated image publishing through GitHub Actions and generated Playwright PDF assessments with production-sizing recommendations.

Achievements

- Received **Best Use of AI** recognition for YB Voyager Copilot at the YugabyteDB internal hackathon.
- Earned **Guardian status with a maximum rating of 2145** on LeetCode. Reached **4-star status with a 1944 rating** on CodeChef and **Specialist status with a maximum rating of 1484** on Codeforces.
- Secured **3rd place** in Algorithmia at Tantrafiesta 2023.